

# Chi-square Tests

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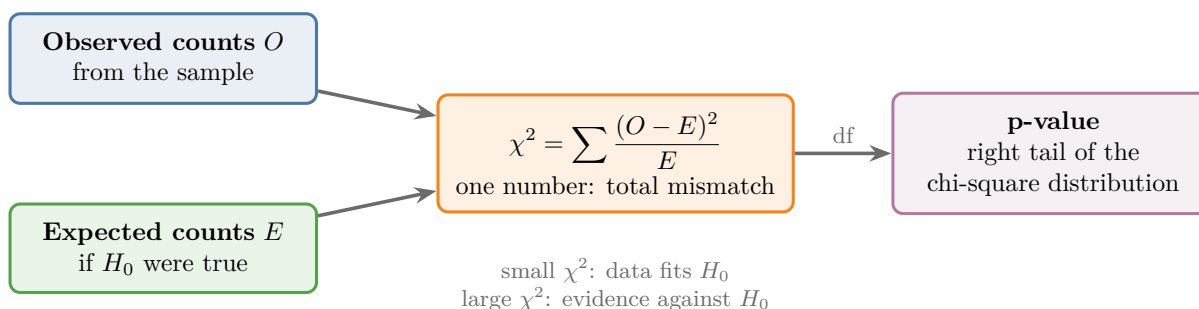
**Where this fits:** Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Hypothesis testing: null and alternative ([Note 291](#)); Joint and marginal probability ([Note 341](#)).
- **Compare with:** One-sample proportion test ([Note 570](#)).

## 1. Overview

**Key point:** A chi-square test compares the counts we observe in each category with the counts we would expect if  $H_0$  were true; the bigger the total mismatch  $\chi^2$ , the smaller the p-value.



**Figure 1:** The chi-square test in four steps

The chi-square test was named in the [what is statistics Note](#) as the test for categorical data. In the [choosing a hypothesis test Note](#) it is the test for two categorical columns. Figure 1 shows how every chi-square test works: observed counts and expected counts go into one number,  $\chi^2$ , and its tail area under the chi-square distribution is the p-value.

This Note covers:

- the chi-square statistic and the chi-square distribution;
- the **goodness-of-fit test**: does one categorical column match claimed proportions?
- the **test of independence**: are two categorical columns related?
- a case study on the Titanic, the conditions the test needs, and its strength measure, Cramér's V.

**Extra:** All of this Note is extra material beyond the “which test when” guide; the guide only names the chi-square test.

## 2. The chi-square statistic

**Key point:**  $\chi^2 = \sum (O - E)^2 / E$  adds up, over all cells, the squared gap between observed and expected counts, each scaled by the expected count.

### 2.1 Observed and expected counts

**Key point:** Observed counts come from the data; expected counts are what  $H_0$  predicts for a sample of the same size.

A categorical column is summarised by counts, one per category (see the [frequency tables Note](#)). For a chi-square test we need two sets of counts:

- **observed counts**  $O$ : how many rows of the sample fall in each category or cell;
- **expected counts**  $E$ : how many would fall there, on average, if  $H_0$  were true.

If  $H_0$  is true,  $O$  and  $E$  differ only by chance. A large mismatch is evidence against  $H_0$ .

### 2.2 The formula

**Key point:** Square each gap so that positive and negative gaps both count, divide by  $E$  so that a gap of 5 matters more in a small cell than in a big one, and add.

1. **In words:** for each cell, take the gap between observed and expected, square it, divide by expected; add over all cells.
2. **Formula:**

$$\chi^2 = \sum_{\text{cells}} \frac{(O - E)^2}{E}$$

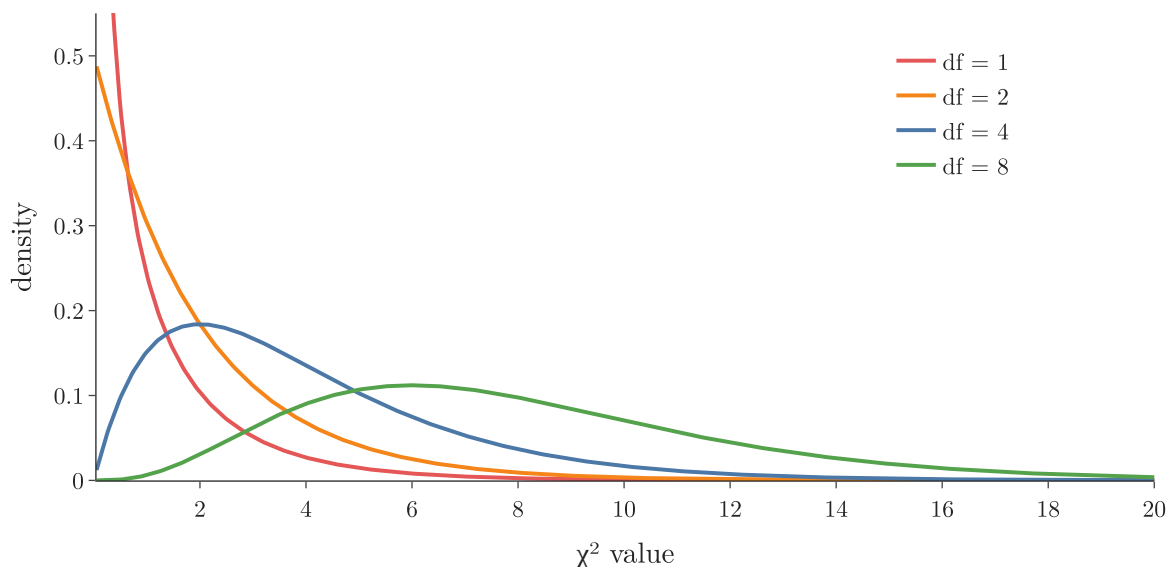
3. **Example:** three cells with  $O = 20, 26, 14$  and  $E = 15, 33, 12$  (section 4):

$$\chi^2 = \frac{(20 - 15)^2}{15} + \frac{(26 - 33)^2}{33} + \frac{(14 - 12)^2}{12} = 1.667 + 1.485 + 0.333 = 3.48$$

$\chi^2$  is 0 only when every observed count equals its expected count. It can never be negative.

## 3. The chi-square distribution

**Key point:** If  $H_0$  is true,  $\chi^2$  follows a chi-square distribution: never negative, skewed to the right, with one parameter, the degrees of freedom.



**Figure 2:** Chi-square densities for four degrees of freedom

The **chi-square distribution** is a continuous distribution on the positive numbers. Like Student's t-distribution (see the [t-procedure Note](#)), it has one parameter, the **degrees of freedom** (df). Figure 2 shows four of them:

- with 1 or 2 df, the density is highest at 0 and falls steadily;
- with more df, the peak moves right and the curve spreads out;
- the mean of a chi-square distribution equals its df.

Only large  $\chi^2$  values count as evidence against  $H_0$ , so the p-value is always the **right-tail** area beyond our  $\chi^2$ . The 5% critical values are 3.84 for 1 df, 5.99 for 2 df and 7.81 for 3 df.

**Extra:** Where the distribution comes from: the sum of the squares of  $k$  independent standard normal variables follows a chi-square distribution with  $k$  df. Each  $(O - E)/\sqrt{E}$  is roughly standard normal for large counts, which is why  $\chi^2$  follows this distribution.

## 4. The goodness-of-fit test

**Key point:** One categorical column against claimed proportions: expected count = sample size  $\times$  claimed proportion, and the degrees of freedom are the number of categories minus 1.

### 4.1 The question

**Key point:** Do the age groups in our sample of 60 match a population that is 25% children, 55% adults and 20% elderly?

The **goodness-of-fit test** asks whether the counts of one categorical column fit a claimed distribution. We use the 60 people of the [choosing a hypothesis test Note](#). Suppose a census says the population is 25% children, 55% adults and 20% elderly.

- $H_0$ : the age groups occur in the claimed proportions 0.25, 0.55, 0.20;
- $H_1$ : at least one proportion differs.

## 4.2 Expected counts and the statistic

**Key point:**  $E = 60 \times 0.25 = 15$  children,  $60 \times 0.55 = 33$  adults,  $60 \times 0.20 = 12$  elderly;  $\chi^2 = 3.48$ .

1. **In words:** the expected count of each category is the sample size times its claimed share.
2. **Formula:**

$$E_i = n \times \pi_i, \quad df = k - 1$$

where  $k$  is the number of categories.

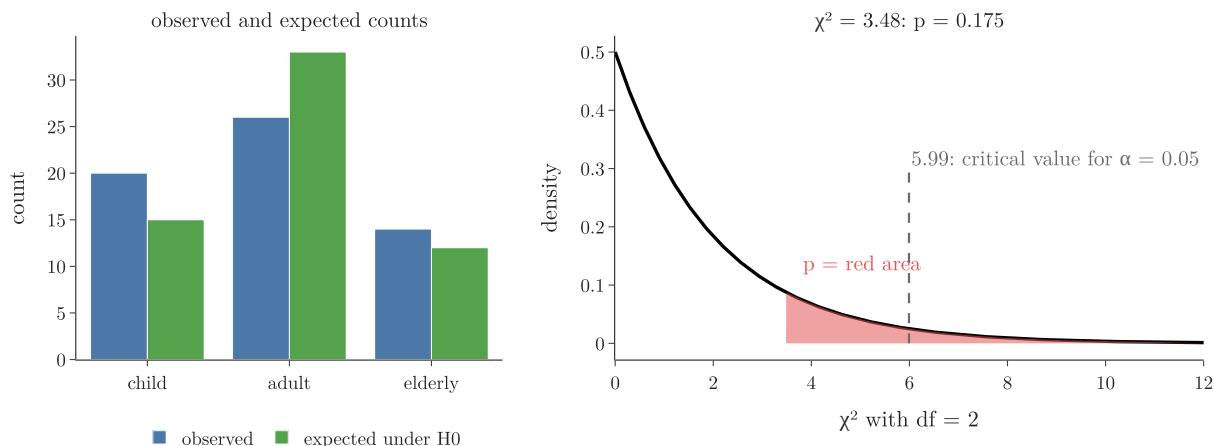
3. **Example:**

|                      | child | adult | elderly | total           |
|----------------------|-------|-------|---------|-----------------|
| observed $O$         | 20    | 26    | 14      | 60              |
| claimed share $\pi$  | 0.25  | 0.55  | 0.20    | 1               |
| expected $E = 60\pi$ | 15    | 33    | 12      | 60              |
| $(O - E)^2/E$        | 1.667 | 1.485 | 0.333   | $\chi^2 = 3.48$ |

The degrees of freedom are  $k - 1 = 2$ : once two expected counts are known, the third is fixed by the total of 60.

## 4.3 The decision

**Key point:**  $\chi^2 = 3.48$  with 2 df gives  $p = 0.175 > 0.05$ : we fail to reject  $H_0$ .



**Figure 3:** Goodness of fit for the age groups: the counts (left) and the p-value as a tail area (right)

Figure 3 shows both halves of the test. On the left, more children and fewer adults were observed than expected. On the right,  $\chi^2 = 3.48$  sits below the critical value 5.99, and the red tail area beyond it is  $p = 0.175$ .

Since  $0.175 > 0.05$ , we fail to reject  $H_0$ : the sample is consistent with the census proportions. The gaps on the left are within what chance produces in 60 people.

**Python:** `chisquare` takes the observed and expected counts.

```
from scipy import stats
```

```
stats.chisquare(f_obs=[20, 26, 14], f_exp=[15, 33, 12])
```

```
# statistic = 3.48, pvalue = 0.175
```

Without `f_exp`, `scipy` assumes equal proportions in all categories.

#### 4.4 Two categories: the proportion test again

**Key point:** With two categories, the goodness-of-fit  $\chi^2$  equals  $z^2$  of the one-sample proportion test, and the p-values are identical.

For gender (26 men, 34 women) against a 50/50 claim, the expected counts are 30 and 30:

$$\chi^2 = \frac{(26 - 30)^2}{30} + \frac{(34 - 30)^2}{30} = 0.533 + 0.533 = 1.067$$

With 1 df this gives  $p = 0.302$ . The proportion test in the [choosing a hypothesis test Note](#) gave  $z = -1.033$ , and  $(-1.033)^2 = 1.067$ , with the same  $p = 0.302$ . The goodness-of-fit test is the proportion test extended to any number of categories.

**Extra:** Goodness of fit also checks whether counts follow a named distribution, such as Poisson (see the [Poisson distribution Note](#)). The expected counts then come from the PMF, and every parameter estimated from the data (such as  $\lambda$  from the mean) removes one more degree of freedom:  $df = k - 1 - (\text{number of estimated parameters})$ .

## 5. The test of independence

**Key point:** Two categorical columns in a contingency table: if they are independent, each expected count is row total  $\times$  column total / grand total, and  $df = (r - 1)(c - 1)$ .

### 5.1 The question and the table

**Key point:** Is the gender mix the same in every age group, or do gender and age group go together?

The **chi-square test of independence** asks whether two categorical columns are related. The counts go in a contingency table (see the [contingency tables Note](#)):

|              | child | adult | elderly | row total |
|--------------|-------|-------|---------|-----------|
| female       | 12    | 12    | 10      | 34        |
| male         | 8     | 14    | 4       | 26        |
| column total | 20    | 26    | 14      | 60        |

- $H_0$ : gender and age group are independent;
- $H_1$ : they are related.

### 5.2 Expected counts under independence

**Key point:** Independence means  $P(A \text{ and } B) = P(A)P(B)$ ; multiplied by  $n$ , this gives  $E = \text{row total} \times \text{column total} / n$ .

Two events are independent when the probability of both is the product of their probabilities (see the [independent events Note](#)). The marginal probabilities come from the totals (see the [joint and marginal probability Note](#)).

1. **In words:** the expected count of a cell is its row total times its column total, divided by the grand total.
2. **Formula:**

$$E = n \times \frac{\text{row total}}{n} \times \frac{\text{column total}}{n} = \frac{\text{row total} \times \text{column total}}{n}$$

3. **Example:** female children:  $P(\text{female}) = 34/60$ ,  $P(\text{child}) = 20/60$ , so

$$E = \frac{34 \times 20}{60} = \frac{680}{60} = 11.33$$

All six expected counts:

|              | child | adult | elderly | row total |
|--------------|-------|-------|---------|-----------|
| female       | 11.33 | 14.73 | 7.93    | 34        |
| male         | 8.67  | 11.27 | 6.07    | 26        |
| column total | 20    | 26    | 14      | 60        |

The expected table has the same totals as the observed one; only the inside is spread out “as if independent”.

### 5.3 The statistic and the decision

**Key point:**  $\chi^2 = 2.50$  with 2 df gives  $p = 0.29$ : no evidence that gender and age group are related.

The contribution  $(O - E)^2/E$  of each cell; for female children it is  $(12 - 11.33)^2/11.33 = 0.039$ :

|        | child | adult | elderly |
|--------|-------|-------|---------|
| female | 0.039 | 0.507 | 0.538   |
| male   | 0.051 | 0.663 | 0.704   |

1. **In words:** add the six contributions; the degrees of freedom are  $(\text{rows} - 1)$  times  $(\text{columns} - 1)$ .
2. **Formula:**

$$\chi^2 = \sum \frac{(O - E)^2}{E}, \quad df = (r - 1)(c - 1)$$

3. **Example:**

$$\chi^2 = 0.039 + 0.507 + 0.538 + 0.051 + 0.663 + 0.704 = 2.50, \quad df = (2 - 1)(3 - 1) = 2$$

With 2 df,  $p = P(\chi^2 \geq 2.50) = 0.29$ . Since  $0.29 > 0.05$ , we fail to reject  $H_0$ . The elderly group has relatively few men (4 observed, 6.07 expected), but in a sample of 60 a gap like this is common by chance.

Why  $(r - 1)(c - 1)$ : with the row and column totals fixed, filling  $(r - 1)(c - 1)$  cells freely determines all the others. In a 2 by 3 table, choosing 2 cells of the female row fixes the rest.

**Python:** `chi2_contingency` takes the observed table and returns the statistic, p-value, df and expected counts.

```
import pandas as pd
```

```
table = pd.crosstab(people["gender"], people["age_group"])
result = stats.chi2_contingency(table)
result.statistic, result.pvalue, result.dof      # 2.50, 0.29, 2
result.expected_freq                             # the expected table
```

## 6. Case study: survival on the Titanic

**Key point:** Sex and class were both strongly related to survival:  $\chi^2 = 263$  ( $p \approx 10^{-59}$ ) for sex,  $\chi^2 = 103$  ( $p \approx 10^{-23}$ ) for class.

### 6.1 Sex and survival

**Key point:** 81 women died where independence predicts 194; 109 men survived where it predicts 222.

The Kaggle Titanic training file has 891 passengers. Survival against sex:

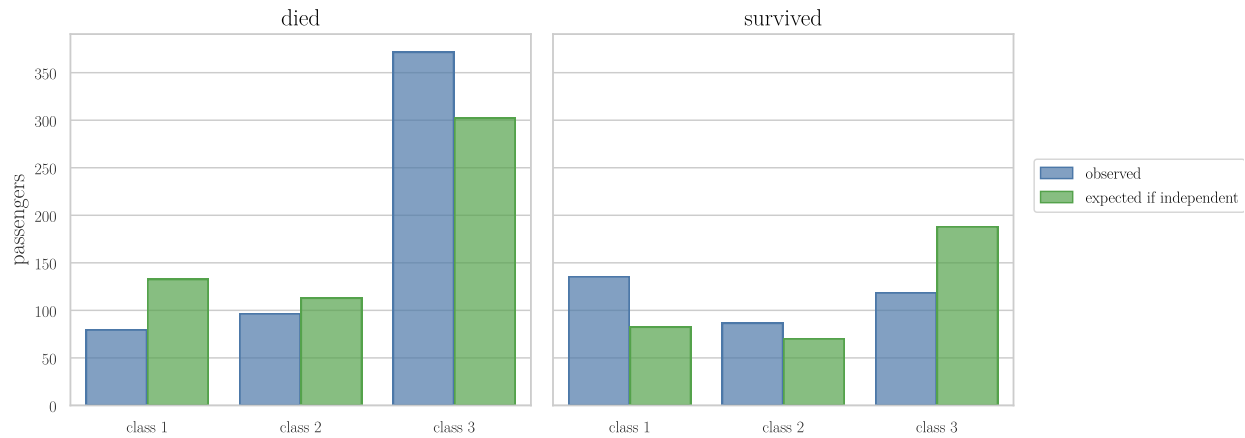
|                  | died  | survived |
|------------------|-------|----------|
| female, observed | 81    | 233      |
| female, expected | 193.5 | 120.5    |
| male, observed   | 468   | 109      |
| male, expected   | 355.5 | 221.5    |

The four contributions are 65.4, 105.0, 35.6 and 57.1, so  $\chi^2 = 263.1$  with  $df = 1$  and  $p = 3.7 \times 10^{-59}$ . We reject  $H_0$ : survival depended on sex.

**Extra:** For a 2 by 2 table, `chi2_contingency` applies **Yates' continuity correction** by default: it shrinks each  $|O - E|$  by 0.5 before squaring, which makes the test slightly more cautious for small counts. Here it gives  $\chi^2 = 260.7$  instead of 263.1; pass `correction=False` to get the plain statistic. With counts this large the choice does not matter.

### 6.2 Class and survival

**Key point:** First class had 136 survivors where independence predicts 83; third class had 119 where it predicts 189.



**Figure 4:** Titanic passengers by class and outcome: observed counts and counts expected under independence

The crosstab of class against survival was drawn as a heatmap in the [bivariate analysis Note](#). Figure 4 compares its observed counts with the expected ones. First class has far more survivors than independence predicts, third class far fewer. The survival rates were 63%, 47% and 24% for the three classes, against 38% overall.

The test gives  $\chi^2 = 102.9$  with  $df = (3 - 1)(2 - 1) = 2$  and  $p = 4.5 \times 10^{-23}$ . We reject  $H_0$ : survival depended on class.

## 7. Conditions and limits

**Key point:** Use raw counts, independent observations, and expected counts of at least 5 in every cell; a significant  $\chi^2$  says that a relationship exists, not how strong it is.

### 7.1 Conditions

**Key point:** Counts, independence, and large enough expected counts.

1. **Counts, not percentages.** The formula needs the number of rows in each cell. Percentages or means give a wrong  $\chi^2$ .
2. **Independent observations.** Each row is counted in exactly one cell, and rows do not influence each other. The same person measured twice breaks this.
3. **Large enough expected counts.** Every expected count should be at least 5; the chi-square distribution is only an approximation, and it fails for small counts. All six expected counts of section 5.2 are above 5 (the smallest is 6.07). For small 2 by 2 tables, **Fisher's exact test** (`stats.fisher_exact`) avoids the approximation; for bigger tables, merging rare categories helps.

## 7.2 Strength of a relationship: Cramér's V

**Key point:** Cramér's V rescales  $\chi^2$  to a number between 0 (no relationship) and 1 (perfect relationship).

$\chi^2$  grows with the sample size: the same pattern in 10 times as many rows gives a 10 times larger  $\chi^2$ . To measure strength, we divide the size out.

1. **In words:** divide  $\chi^2$  by the sample size and by one less than the smaller table dimension, then take the square root.
2. **Formula:**

$$V = \sqrt{\frac{\chi^2}{n \times (\min(r, c) - 1)}}$$

3. **Example:** Titanic sex and survival,  $\chi^2 = 263.1$ ,  $n = 891$ , a 2 by 2 table:

$$V = \sqrt{\frac{263.1}{891 \times 1}} = \sqrt{0.295} = 0.54$$

For class and survival,  $V = \sqrt{102.9/891} = 0.34$ . Both relationships are significant, but sex was the stronger predictor of survival. In scipy: `stats.contingency.association(table, method="cramer")`.

## 7.3 Chi-square in machine learning

**Key point:** Feature selection with `SelectKBest(chi2)` ranks categorical features by how strongly they are related to the target.

Feature selection with `SelectKBest(score_func=chi2)` (see the [pipelines Note](#)) scores each feature against the target with a chi-square statistic: a feature whose counts differ strongly between the classes gets a high score. scikit-learn's version treats each feature's values as counts, which is why it needs values of 0 or more; for one-hot encoded columns this matches the test of independence above.

## 8. Summary

|                    | Goodness of fit                                     | Test of independence                              |
|--------------------|-----------------------------------------------------|---------------------------------------------------|
| Columns            | one categorical                                     | two categorical                                   |
| $H_0$              | the categories have the claimed proportions         | the two columns are independent                   |
| Expected count     | $n \times \pi_i$                                    | row total $\times$ column total $/n$              |
| Degrees of freedom | $k - 1$                                             | $(r - 1)(c - 1)$                                  |
| scipy              | <code>chisquare</code>                              | <code>chi2_contingency</code>                     |
| Example here       | age groups vs census:<br>$\chi^2 = 3.48, p = 0.175$ | gender vs age group:<br>$\chi^2 = 2.50, p = 0.29$ |

- Under  $H_0$ ,  $\chi^2 = \sum(O - E)^2/E$  follows a chi-square distribution; the p-value is the right-tail area.
- With two categories, goodness of fit is the one-sample proportion test:  $\chi^2 = z^2$ .
- The test needs raw counts, independent rows and expected counts of at least 5.

## 9. Key terms

| Term                            | Meaning                                                                                                           |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Observed count $O$              | The number of sample rows in a category or table cell                                                             |
| Expected count $E$              | The number of rows a category or cell would hold on average if $H_0$ were true                                    |
| Chi-square statistic $\chi^2$   | $\sum(O - E)^2/E$ : the total mismatch between observed and expected counts                                       |
| Chi-square distribution         | The distribution of $\chi^2$ under $H_0$ : positive, right-skewed, with one parameter, the degrees of freedom     |
| Goodness-of-fit test            | A chi-square test of whether one categorical column follows claimed proportions;<br>$df = k - 1$                  |
| Chi-square test of independence | A chi-square test of whether two categorical columns are related; $df = (r - 1)(c - 1)$                           |
| Yates' continuity correction    | A small adjustment to $\chi^2$ for 2 by 2 tables, applied by default in <code>chi2_contingency</code>             |
| Fisher's exact test             | An exact test for small 2 by 2 tables, used when expected counts fall below 5                                     |
| Cramér's V                      | $\sqrt{\chi^2/(n(\min(r, c) - 1))}$ : the strength of a relationship between two categorical columns, from 0 to 1 |